ClipSAT : Math for AI & Data Science

Unit 8.01: PCA & Dimensionality Reduction

Name: _____

Q1. Principal Component Analysis PCA projects a dataset from mathbb{R}^{50} onto its top 6 principal components. What is the dimension of each transformed data point?

[ ] A: 25
[ ] B: 50
[ ] C: 44
[ ] D: 6

Q2. A 3D dataset has covariance matrix eigenvalues lambda_1 = 62 , lambda_2 = 24 , and lambda_3 = 14 . What percentage of total variance is explained by the first principal component PC_1 ?

[ ] A: 62.0%
[ ] B: 86.0%
[ ] C: 24.0%
[ ] D: 72.1%

Q3. A 3D dataset has covariance matrix eigenvalues lambda_1 = 64 , lambda_2 = 23 , and lambda_3 = 13 . What percentage of total variance is explained by the first principal component PC_1 ?

[ ] A: 64.0%
[ ] B: 87.0%
[ ] C: 23.0%
[ ] D: 73.6%

Q4. A 3D dataset has covariance matrix eigenvalues lambda_1 = 66 , lambda_2 = 22 , and lambda_3 = 12 . What percentage of total variance is explained by the first principal component PC_1 ?

[ ] A: 66.0%
[ ] B: 88.0%
[ ] C: 22.0%
[ ] D: 75.0%

Q5. A 3D dataset has covariance matrix eigenvalues lambda_1 = 68 , lambda_2 = 21 , and lambda_3 = 11 . What percentage of total variance is explained by the first principal component PC_1 ?

[ ] A: 68.0%
[ ] B: 89.0%
[ ] C: 21.0%
[ ] D: 76.4%

Q6. A 3D dataset has covariance matrix eigenvalues lambda_1 = 70 , lambda_2 = 20 , and lambda_3 = 10 . What percentage of total variance is explained by the first principal component PC_1 ?

[ ] A: 70.0%
[ ] B: 90.0%
[ ] C: 20.0%
[ ] D: 77.8%

Q7. A 3D dataset has covariance matrix eigenvalues lambda_1 = 72 , lambda_2 = 19 , and lambda_3 = 9 . What percentage of total variance is explained by the first principal component PC_1 ?

[ ] A: 72.0%
[ ] B: 91.0%
[ ] C: 19.0%
[ ] D: 79.1%

Q8. A 3D dataset has covariance matrix eigenvalues lambda_1 = 74 , lambda_2 = 18 , and lambda_3 = 8 . What percentage of total variance is explained by the first principal component PC_1 ?

[ ] A: 74.0%
[ ] B: 92.0%
[ ] C: 18.0%
[ ] D: 80.4%